CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

M & M CARDOSO & SONS DAIRY

16489 AUGUST AVE

DELHI CA 95315

NOTES:

PREPARED BY:

MARIANN PEDROSO
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: M & M Cardoso & Sons Dairy LTD

Physical address of dairy:

Number and Street	City	County	Zip Code
16489 August AVE	Delhi	Merced	95315

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 04/13/1977

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X046-X190-X010-XXXX

B. OPERATORS

Cardoso, Armelim or Tiberio

Operator name: Cardoso, Armelim or Tiberio

Telephone no.: (209) 648-4032 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
16489 August AVE	Delhi	CA	95315

C. OWNERS

M & M Cardoso and Sons LTD

Legal owner name: M & M Cardoso and Sons LTD

Telephone no.: (209) 648-4032 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
16489 August AVE	Delhi	CA	95315

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	72	245	246	0	0
Number under roof	870	0	0	0	0	0
Maximum number	870	72	245	246	0	0
Average number	870	72	245	246	0	0
Avg live weight (lbs)	1,000	1,050	925	665		

Predominant milk cow breed: Jersey

Average milk production: 62 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 18,246.06 *tons per reporting period*

Total nitrogen from manure: 228,360.62 *lbs per reporting period*

After ammonia losses (30% loss applied): 159,852.43 *lbs per reporting period*

Total phosphorus from manure: 38,225.46 *lbs per reporting period*

Total potassium from manure: 108,567.00 *lbs per reporting period*

Total salt from manure: 426,195.90 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 34,431,000 *gallons*

Total nitrogen generated: 60,328.07 *lbs*

Total phosphorus generated: 16,428.87 *lbs*

Total potassium generated: 92,716.12 *lbs*

Total salt generated: 513,537.45 *lbs*

	34,431,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	34,431,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	Corral solids	1,274.00 *ton*	Dry-weight	26.0		5,900.00	3,400.00	21,800.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	11,124.57	6,410.77	41,104.34	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	11,124.57	6,410.77	41,104.34	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Armelin	28	28	2	manure	X046-X190-X046-XXXX X046-X190-X047-XXXX
Denair	20	19	6	manure	X019-X034-X013-XXXX
Home	61	61	2	process wastewater	X046-X190-X010-XXXX
Horseranch	33	33	2	process wastewater	X046-X190-X038-XXXX
River	38	38	2	manure	X047-X050-X025-XXXX
Vineyard	20	20	2	process wastewater	X046-X190-X027-XXXX
Totals for areas that were used for application	200	199	16		
Totals for areas that were not used for application					
Land application area totals	200	199	16		

B. CROPS AND HARVESTS

Armelin

Field name: Armelin

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	336.00 *ton*	Dry-weight		80.8	15,400.00	3,600.00	18,000.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.00	70.96	16.59	82.94	304.13

Armelin

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/05/2023	858.00 *ton*	Dry-weight		72.7	13,400.00	4,100.00	19,100.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.64	224.20	68.60	319.56	1,288.29

Denair

Field name: Denair

10/05/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 19 Plant date: 10/05/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/15/2023	39.00 *ton*	Dry-weight		23.8	35,300.00	3,200.00	31,400.00		12.40
05/14/2023	52.00 *ton*	Dry-weight		23.8	35,300.00	3,200.00	31,400.00		12.40
06/14/2023	104.00 *ton*	Dry-weight		23.8	35,300.00	3,200.00	31,400.00		12.40
07/18/2023	35.00 *ton*	Dry-weight		6.9	29,100.00	3,500.00	25,800.00		10.00
08/22/2023	30.80 *ton*	Dry-weight		6.9	29,100.00	3,500.00	25,800.00		10.00
09/23/2023	20.30 *ton*	Dry-weight		6.9	29,100.00	3,500.00	25,800.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	14.79	797.67	79.58	708.82	2,783.27

Home

Field name: Home

Home

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	996.00 *ton*	Dry-weight		74.7	29,600.00	3,300.00	42,100.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.33	244.55	27.26	347.83	1,123.62

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/07/2023	1,914.30 *ton*	Dry-weight		73.0	13,900.00	3,400.00	21,100.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.38	235.55	57.62	357.57	1,321.81

Horseranch

Field name: Horseranch

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	516.00 *ton*	Dry-weight		75.2	25,500.00	4,800.00	49,600.00		17.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.64	197.77	37.23	384.68	1,326.21

Horseranch

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	1,066.00 *ton*	Dry-weight		68.3	22,000.00	3,000.00	15,900.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.30	450.56	61.44	325.63	1,290.25

River

Field name: River

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	324.00 *ton*	Dry-weight		45.1	12,300.00	2,300.00	15,200.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.05	230.30	43.06	284.60	1,460.46

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	1,053.00 *ton*	Dry-weight		74.0	15,700.00	0.30	17,900.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.71	226.23	0.00	257.93	907.80

Vineyard

Field name: Vineyard

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	390.00 *ton*	Dry-weight		77.5	33,500.00	4,600.00	51,600.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.50	293.96	40.37	452.79	1,272.38

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	637.00 *ton*	Dry-weight		74.9	15,500.00	3,400.00	23,600.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.85	247.82	54.36	377.33	1,311.07

NUTRIENT BUDGET

A. LAND APPLICATIONS

Armelin - 11/05/2022: Oats, silage-soft dough

Field name: Armelin

Crop: Oats, silage-soft dough — Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	76.74	11.23	9.09	0.00	196.00 *ton*
Application event totals		76.74	11.23	9.09	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.04	0.00	0.00	0.00	4,739,336.00 *gal*
Application event totals		9.04	0.00	0.00	0.00	

Armelin - 05/21/2023: Corn, silage

Field name: Armelin

Crop: Corn, silage — Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	198.82	65.66	45.60	0.00	448.00 *ton*
Application event totals		198.82	65.66	45.60	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.18	0.00	0.00	0.00	5,335,276.00 *gal*
Application event totals		10.18	0.00	0.00	0.00	

Armelin - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.35	0.00	0.00	0.00	4,376,511.00 *gal*
Application event totals		8.35	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	16.63	0.00	0.00	0.00	8,718,053.00 *gal*
Application event totals		16.63	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.86	0.00	0.00	0.00	8,315,860.00 *gal*
Application event totals		15.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.65	0.00	0.00	0.00	8,206,586.00 *gal*
Application event totals		15.65	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	14.62	0.00	0.00	0.00	7,665,266.00 *gal*
Application event totals		14.62	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	14.47	0.00	0.00	0.00	7,585,811.00 *gal*
Application event totals		14.47	0.00	0.00	0.00	

Denair - 10/05/2020: Alfalfa, hay

Denair - 10/05/2020: Alfalfa, hay

Field name: Denair

Crop: Alfalfa, hay

Plant date: 10/05/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.83	0.00	0.00	0.00	2,431,214.00 *gal*
Application event totals		6.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.08	0.00	0.00	0.00	2,161,988.00 *gal*
Application event totals		6.08	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.99	0.00	0.00	0.00	2,129,822.00 *gal*
Application event totals		5.99	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.24	0.00	0.00	0.00	2,218,238.00 *gal*
Application event totals		6.24	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	0.00	2,712,498.00 *gal*
Application event totals		7.62	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.27	0.00	0.00	0.00	2,230,297.00 *gal*
Application event totals		6.27	0.00	0.00	0.00	

Denair - 10/05/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.78	0.00	0.00	0.00	2,411,106.00 *gal*
Application event totals		6.78	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.93	0.00	0.00	0.00	2,109,746.00 *gal*
Application event totals		5.93	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.59	0.00	0.00	0.00	2,342,797.00 *gal*
Application event totals		6.59	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.93	0.00	0.00	0.00	2,109,746.00 *gal*
Application event totals		5.93	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.07	0.00	0.00	0.00	2,157,947.00 *gal*
Application event totals		6.07	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	323.08	106.70	74.10	0.00	494.00 *ton*
Application event totals		323.08	106.70	74.10	0.00	

Home - 11/18/2022: Oats, silage-soft dough

Home - 11/18/2022: Oats, silage-soft dough

Field name: Home

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	65.67	28.88	86.09	552.21	2,070,000.00 *gal*
Irrigation Well	Ground water	9.08	0.00	0.00	0.00	5,184,000.00 *gal*
Canal	Surface water	2.91	0.00	0.00	0.00	3,318,509.00 *gal*
Application event totals		77.65	28.88	86.09	552.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.16	21.18	63.13	404.95	1,518,000.00 *gal*
Irrigation Well	Ground water	9.08	0.00	0.00	0.00	5,184,000.00 *gal*
Canal	Surface water	3.14	0.00	0.00	0.00	3,587,605.00 *gal*
Application event totals		60.38	21.18	63.13	404.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	81.96	16.61	138.34	589.02	1,104,000.00 *gal*
Application event totals		81.96	16.61	138.34	589.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	94.49	17.18	129.89	626.78	1,380,000.00 *gal*
Application event totals		94.49	17.18	129.89	626.78	

Home - 05/21/2023: Corn, silage

Field name: Home

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Home - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	103.94	18.90	142.88	689.46	1,518,000.00 *gal*
Irrigation Well	Ground water	9.08	0.00	0.00	0.00	5,184,000.00 *gal*
Canal	Surface water	4.16	0.00	0.00	0.00	4,753,577.00 *gal*
Application event totals		117.18	18.90	142.88	689.46	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	56.69	10.31	77.93	376.07	828,000.00 *gal*
Canal	Surface water	5.89	0.00	0.00	0.00	6,729,672.00 *gal*
Application event totals		62.59	10.31	77.93	376.07	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.75	0.00	0.00	0.00	7,713,336.00 *gal*
Application event totals		6.75	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.22	8.36	45.20	326.91	2,484,000.00 *gal*
Canal	Surface water	7.46	0.00	0.00	0.00	8,520,525.00 *gal*
Application event totals		29.68	8.36	45.20	326.91	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.22	8.36	45.20	326.91	2,484,000.00 *gal*
Canal	Surface water	6.28	0.00	0.00	0.00	7,175,210.00 *gal*
Application event totals		28.51	8.36	45.20	326.91	

Home - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	24.69	9.29	50.22	363.23	2,760,000.00 *gal*
Canal	Surface water	6.28	0.00	0.00	0.00	7,175,210.00 *gal*
Application event totals		30.98	9.29	50.22	363.23	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	24.69	9.29	50.22	363.23	2,760,000.00 *gal*
Irrigation Well	Ground water	9.08	0.00	0.00	0.00	5,184,000.00 *gal*
Canal	Surface water	5.89	0.00	0.00	0.00	6,726,739.00 *gal*
Application event totals		39.66	9.29	50.22	363.23	

Horseranch - 11/17/2022: Oats, silage-soft dough

Field name: Horseranch

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	97.11	42.71	127.31	816.60	1,656,000.00 *gal*
Irrigation Well	Ground water	6.99	0.00	0.00	0.00	2,160,000.00 *gal*
Canal	Surface water	3.60	0.00	0.00	0.00	2,222,051.00 *gal*
Application event totals		107.70	42.71	127.31	816.60	

03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	56.82	11.52	95.90	408.30	414,000.00 *gal*
Application event totals		56.82	11.52	95.90	408.30	

Horseranch - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	94.69	19.19	159.83	680.50	690,000.00 *gal*
Application event totals		94.69	19.19	159.83	680.50	

Horseranch - 05/21/2023: Corn, silage

Field name: Horseranch

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	104.80	19.05	144.06	695.15	828,000.00 *gal*
Canal	Surface water	6.35	0.00	0.00	0.00	3,925,628.00 *gal*
Application event totals		111.15	19.05	144.06	695.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	52.40	9.53	72.03	347.58	414,000.00 *gal*
Canal	Surface water	5.99	0.00	0.00	0.00	3,703,397.00 *gal*
Application event totals		58.39	9.53	72.03	347.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.35	0.00	0.00	0.00	5,777,327.00 *gal*
Application event totals		9.35	0.00	0.00	0.00	

Horseranch - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.54	7.73	41.77	302.14	1,242,000.00 *gal*
Canal	Surface water	11.39	0.00	0.00	0.00	7,036,474.00 *gal*
Application event totals		31.93	7.73	41.77	302.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	20.54	7.73	41.77	302.14	1,242,000.00 *gal*
Canal	Surface water	9.59	0.00	0.00	0.00	5,925,449.00 *gal*
Application event totals		30.13	7.73	41.77	302.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.82	8.58	46.41	335.71	1,380,000.00 *gal*
Canal	Surface water	9.59	0.00	0.00	0.00	5,925,449.00 *gal*
Application event totals		32.41	8.58	46.41	335.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.82	8.58	46.41	335.71	1,380,000.00 *gal*
Canal	Surface water	11.54	0.00	0.00	0.00	7,129,063.00 *gal*
Application event totals		34.36	8.58	46.41	335.71	

River - 11/19/2022: Oats, silage-soft dough

Field name: River

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

River - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	285.05	41.71	33.77	0.00	494.00 *ton*
Application event totals		285.05	41.71	33.77	0.00	

River - 06/21/2023: Corn, silage

Field name: River

Crop: Corn, silage

Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	273.37	90.29	62.70	0.00	836.00 *ton*
Application event totals		273.37	90.29	62.70	0.00	

05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.01	0.00	0.00	385.61	6,552,000.00 *gal*
Application event totals		2.01	0.00	0.00	385.61	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.86	0.00	0.00	355.95	6,048,000.00 *gal*
Application event totals		1.86	0.00	0.00	355.95	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.96	0.00	0.00	375.73	6,384,000.00 *gal*
Application event totals		1.96	0.00	0.00	375.73	

River - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.12	0.00	0.00	405.39	6,888,000.00 *gal*
Application event totals		2.12	0.00	0.00	405.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.07	0.00	0.00	395.50	6,720,000.00 *gal*
Application event totals		2.07	0.00	0.00	395.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.96	0.00	0.00	375.73	6,384,000.00 *gal*
Application event totals		1.96	0.00	0.00	375.73	

Vineyard - 11/16/2022: Oats, silage-soft dough

Field name: Vineyard

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	106.82	46.99	140.04	898.26	1,104,000.00 *gal*
Domestic Well	Ground water	13.84	0.00	0.00	0.00	2,592,000.00 *gal*
Canal	Surface water	2.26	0.00	0.00	0.00	847,340.00 *gal*
Application event totals		122.93	46.99	140.04	898.26	

Vineyard - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	78.12	15.83	131.86	561.41	345,000.00 *gal*
Application event totals		78.12	15.83	131.86	561.41	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	86.46	15.72	118.85	573.50	414,000.00 *gal*
Application event totals		86.46	15.72	118.85	573.50	

Vineyard - 05/21/2023: Corn, silage

Field name: Vineyard

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	115.28	20.96	158.46	764.67	552,000.00 *gal*
Irrigation Well	Ground water	9.23	0.00	0.00	0.00	1,728,000.00 *gal*
Canal	Surface water	5.16	0.00	0.00	0.00	1,932,783.00 *gal*
Application event totals		129.67	20.96	158.46	764.67	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	57.64	10.48	79.23	382.33	276,000.00 *gal*
Canal	Surface water	8.60	0.00	0.00	0.00	3,221,261.00 *gal*
Application event totals		66.24	10.48	79.23	382.33	

Vineyard - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.67	0.00	0.00	0.00	2,496,492.00 *gal*
Application event totals		6.67	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.59	8.50	45.95	332.35	828,000.00 *gal*
Canal	Surface water	7.53	0.00	0.00	0.00	2,818,611.00 *gal*
Application event totals		30.12	8.50	45.95	332.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.59	8.50	45.95	332.35	828,000.00 *gal*
Canal	Surface water	8.39	0.00	0.00	0.00	3,140,763.00 *gal*
Application event totals		30.98	8.50	45.95	332.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	26.36	9.92	53.61	387.75	966,000.00 *gal*
Canal	Surface water	6.45	0.00	0.00	0.00	2,415,962.00 *gal*
Application event totals		32.81	9.92	53.61	387.75	

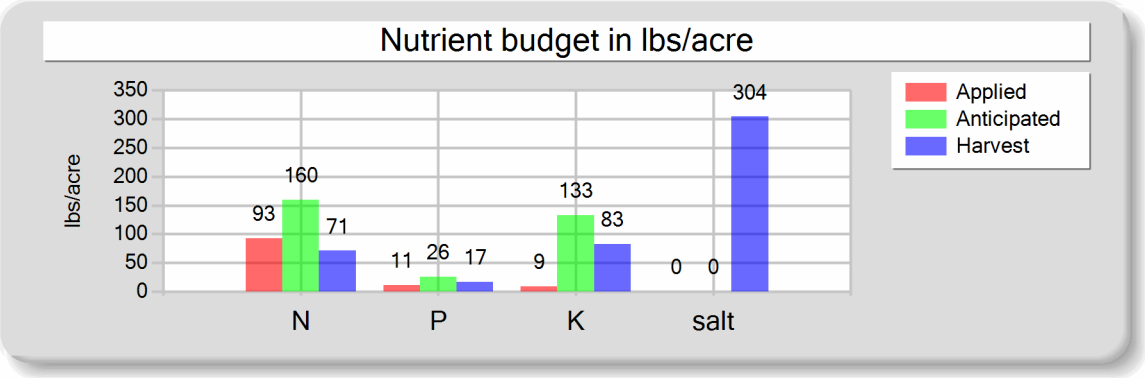
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	26.36	9.92	53.61	387.75	966,000.00 *gal*
Canal	Surface water	10.32	0.00	0.00	0.00	3,865,532.00 *gal*
Application event totals		36.68	9.92	53.61	387.75	

B. NUTRIENT BUDGET

Armelin - 11/05/2022: Oats, silage-soft dough

Field name: Armelin | Crop: Oats, silage-soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	76.74	11.23	9.09	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.04	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.78	11.23	9.09	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	70.96	16.59	82.94	304.13
Nutrient balance	21.82	-5.36	-73.85	-304.13
Applied to removed ratio	1.31	0.68	0.11	0.00

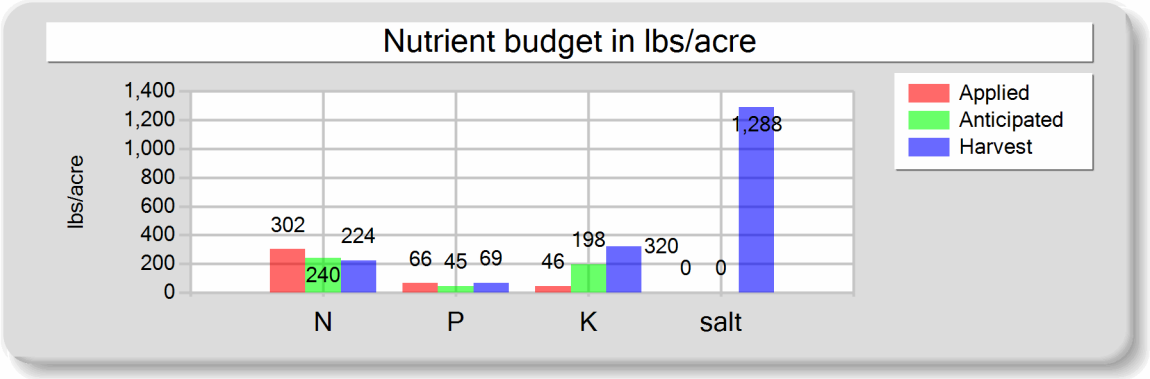
Fresh water applied	
4,739,336.00	*gallons*
174.53	*acre-inches*
6.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Armelin - 05/21/2023: Corn, silage

Field name: Armelin | Crop: Corn, silage | Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	198.82	65.66	45.60	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	95.76	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.58	65.66	45.60	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.20	68.60	319.56	1,288.29
Nutrient balance	77.38	-2.93	-273.96	-1,288.29
Applied to removed ratio	1.35	0.96	0.14	0.00

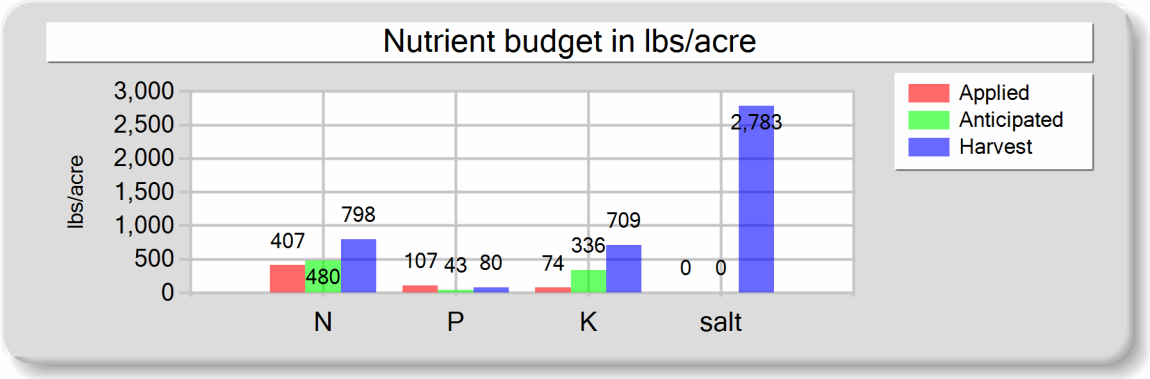
Fresh water applied	
50,203,363.00	*gallons*
1,848.82	*acre-inches*
66.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Denair - 10/05/2020: Alfalfa, hay

Field name: Denair | Crop: Alfalfa, hay | Plant date: 10/05/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	323.08	106.70	74.10	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	70.32	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	407.39	106.70	74.10	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	797.67	79.58	708.82	2,783.27
Nutrient balance	-390.28	27.12	-634.72	-2,783.27
Applied to removed ratio	0.51	1.34	0.10	0.00

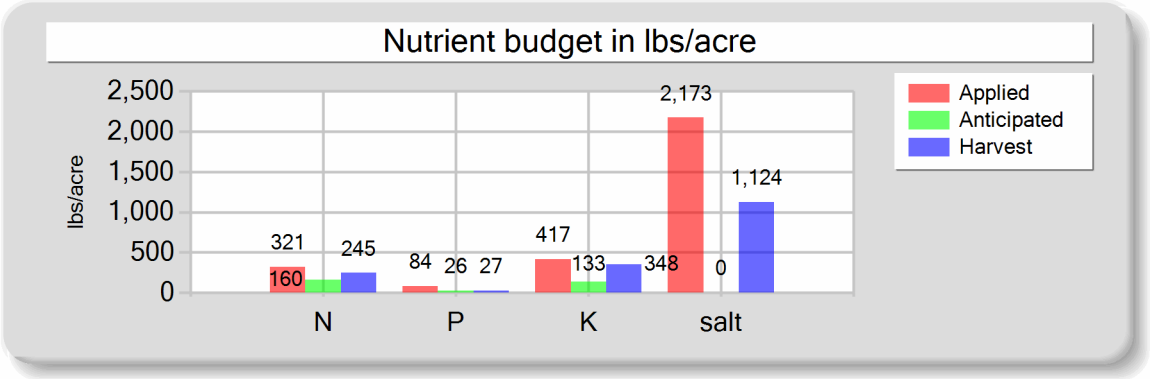
Fresh water applied	
25,015,399.00	*gallons*
921.23	*acre-inches*
48.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

Home - 11/18/2022: Oats, silage-soft dough

Field name: Home | Crop: Oats, silage-soft dough | Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.28	83.86	417.45	2,172.96
Fresh water	24.20	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	321.48	83.86	417.45	2,172.96
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	244.55	27.26	347.83	1,123.62
Nutrient balance	76.93	56.60	69.62	1,049.34
Applied to removed ratio	1.31	3.08	1.20	1.93

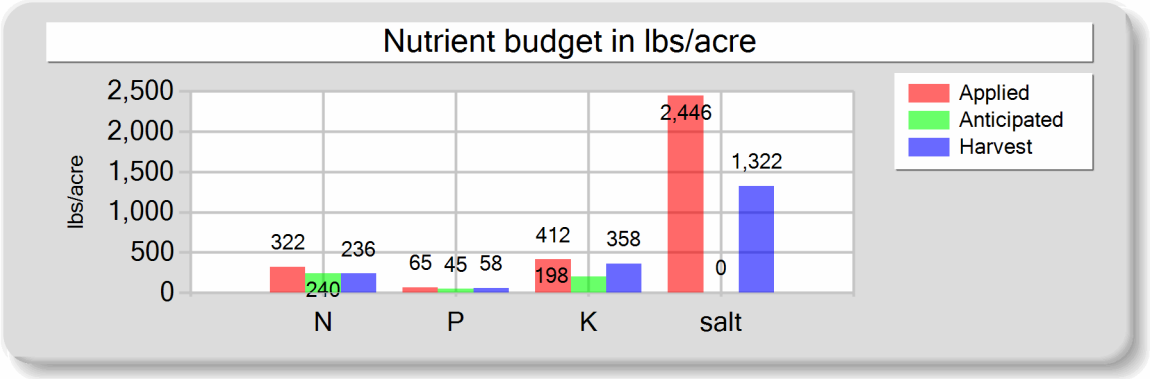
Fresh water applied	
17,274,114.00	*gallons*
636.15	*acre-inches*
10.43	*inches/acre*

Process wastewater applied	
6,072,000.00	*gallons*
223.61	*acre-inches*
3.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 05/21/2023: Corn, silage

Field name: Home | Crop: Corn, silage | Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	254.47	64.50	411.63	2,445.79
Fresh water	60.88	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.34	64.50	411.63	2,445.79
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	235.55	57.62	357.57	1,321.81
Nutrient balance	86.79	6.88	54.07	1,123.98
Applied to removed ratio	1.37	1.12	1.15	1.85

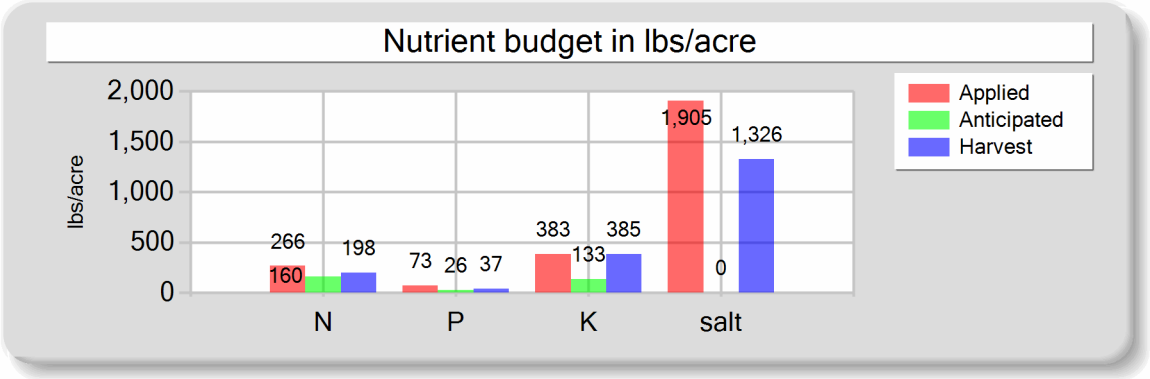
Fresh water applied	
59,162,269.00	*gallons*
2,178.75	*acre-inches*
35.72	*inches/acre*

Process wastewater applied	
12,834,000.00	*gallons*
472.63	*acre-inches*
7.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

Horseranch - 11/17/2022: Oats, silage-soft dough

Field name: Horseranch | Crop: Oats, silage-soft dough | Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.62	73.42	383.03	1,905.39
Fresh water	10.59	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.21	73.42	383.03	1,905.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	197.77	37.23	384.68	1,326.21
Nutrient balance	68.44	36.20	-1.65	579.18
Applied to removed ratio	1.35	1.97	1.00	1.44

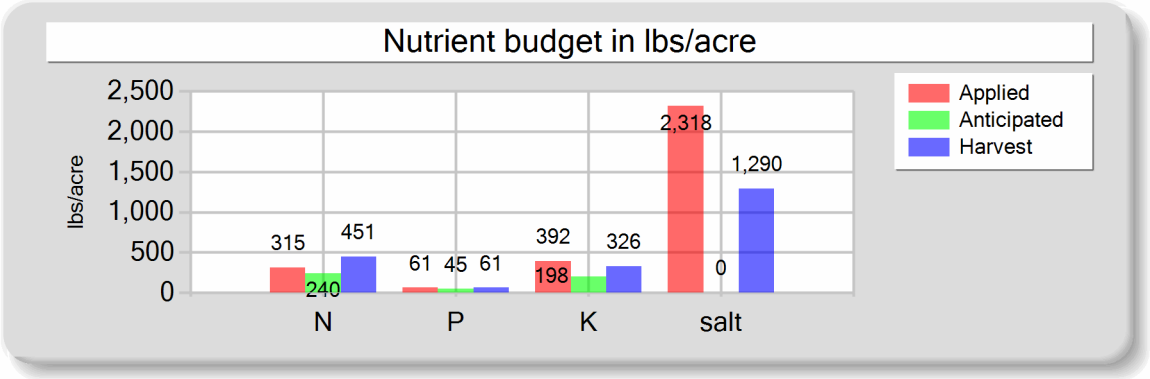
Fresh water applied	
4,382,051.00	*gallons*
161.38	*acre-inches*
4.89	*inches/acre*

Process wastewater applied	
2,760,000.00	*gallons*
101.64	*acre-inches*
3.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

Horseranch - 05/21/2023: Corn, silage

Field name: Horseranch | Crop: Corn, silage | Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	243.92	61.20	392.45	2,318.44
Fresh water	63.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.72	61.20	392.45	2,318.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	450.56	61.44	325.63	1,290.25
Nutrient balance	-135.84	-0.24	66.82	1,028.19
Applied to removed ratio	0.70	1.00	1.21	1.80

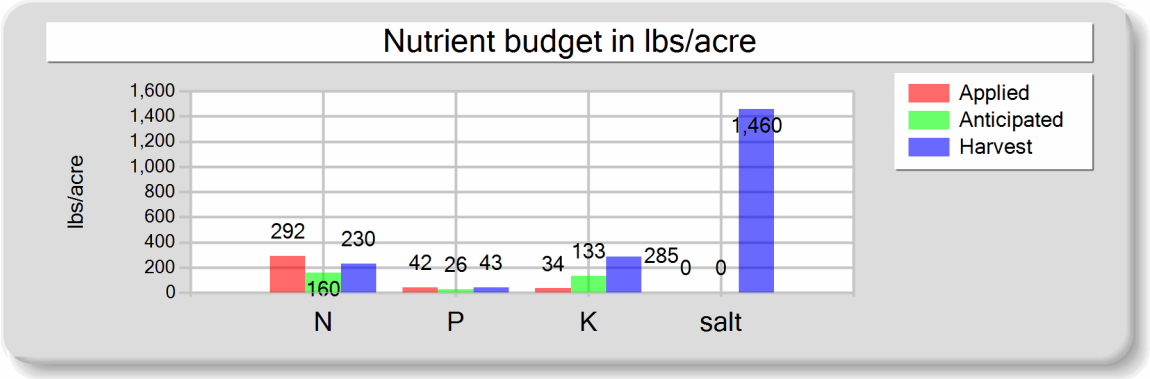
Fresh water applied	
39,422,787.00	*gallons*
1,451.81	*acre-inches*
43.99	*inches/acre*

Process wastewater applied	
6,486,000.00	*gallons*
238.86	*acre-inches*
7.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

River - 11/19/2022: Oats, silage-soft dough

Field name: River | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	285.05	41.71	33.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.05	41.71	33.77	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	230.30	43.06	284.60	1,460.46
Nutrient balance	61.75	-1.35	-250.83	-1,460.46
Applied to removed ratio	1.27	0.97	0.12	0.00

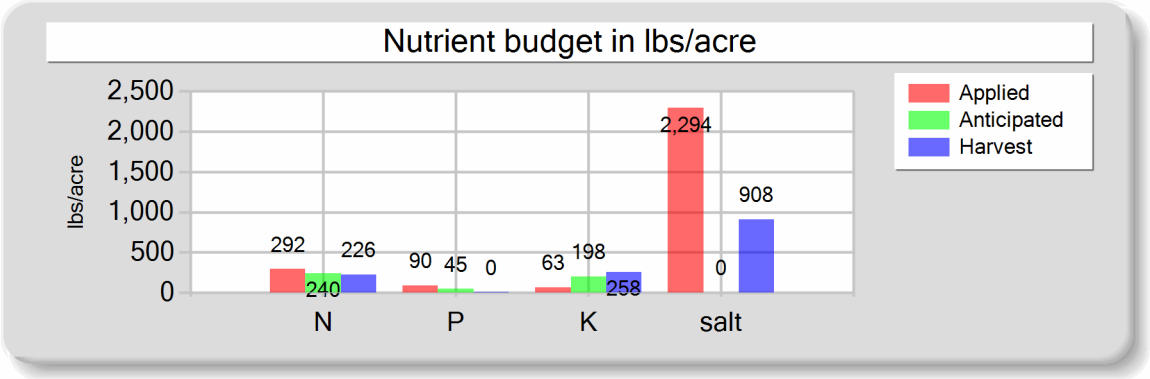
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

River - 06/21/2023: Corn, silage

Field name: River | Crop: Corn, silage | Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	273.37	90.29	62.70	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.98	0.00	0.00	2,293.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.36	90.29	62.70	2,293.90
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	226.23	0.00	257.93	907.80
Nutrient balance	66.13	90.28	-195.23	1,386.10
Applied to removed ratio	1.29	20,886.26	0.24	2.53

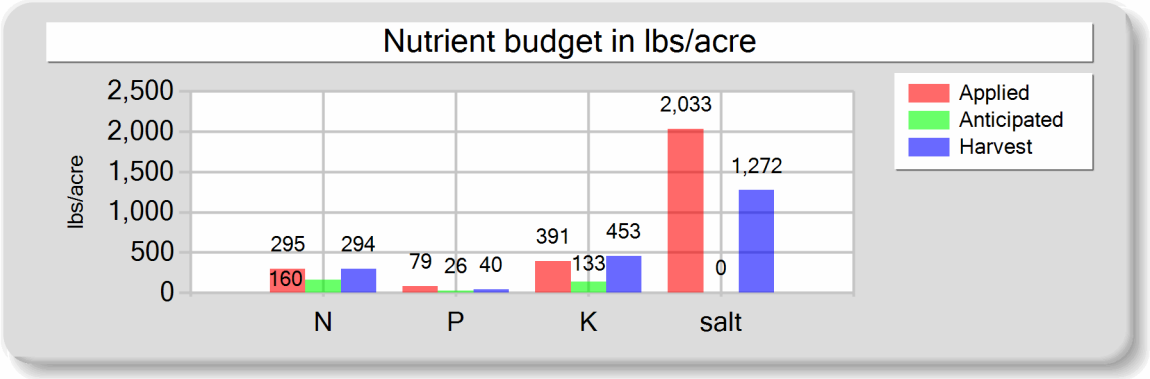
Fresh water applied	
38,976,000.00	*gallons*
1,435.35	*acre-inches*
37.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Vineyard - 11/16/2022: Oats, silage-soft dough

Field name: Vineyard | Crop: Oats, silage-soft dough | Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	271.40	78.54	390.74	2,033.17
Fresh water	16.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.51	78.54	390.74	2,033.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	293.96	40.37	452.79	1,272.38
Nutrient balance	0.55	38.17	-62.05	760.79
Applied to removed ratio	1.00	1.95	0.86	1.60

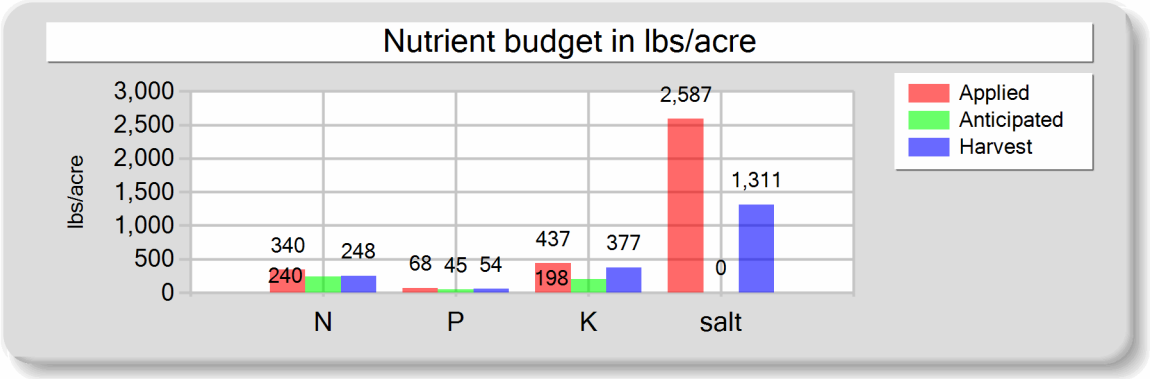
Fresh water applied	
3,439,340.00	*gallons*
126.66	*acre-inches*
6.33	*inches/acre*

Process wastewater applied	
1,863,000.00	*gallons*
68.61	*acre-inches*
3.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Vineyard - 05/21/2023: Corn, silage

Field name: Vineyard | Crop: Corn, silage | Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	270.82	68.27	436.81	2,587.21
Fresh water	62.35	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	340.17	68.27	436.81	2,587.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.82	54.36	377.33	1,311.07
Nutrient balance	92.35	13.91	59.47	1,276.13
Applied to removed ratio	1.37	1.26	1.16	1.97

Fresh water applied	
21,619,404.00	*gallons*
796.17	*acre-inches*
39.81	*inches/acre*

Process wastewater applied	
4,416,000.00	*gallons*
162.63	*acre-inches*
8.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022 SSB Manure

Sample and source description: Fall 2022 SSB Manure

Sample date: 09/07/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,700.00	4,200.00	3,400.00							
DL	0.01	0.01	0.01							

SSB Manure

Sample and source description: SSB Manure

Sample date: 09/28/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,800.00	7,200.00	5,000.00							
DL	0.01	0.01	0.01							

Export Corral Manure

Sample and source description: Export Corral Manure

Sample date: 11/14/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 26.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	3,400.00	21,800.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	231.00	14.40	0.00	0.90	102.00	304.00								3,770.00	1,950
DL	0.07	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	541.00	377.00	0.00	1.70	110.00	916.00	273.00	121.00	214.00	2,970.00	0.00	4.60	404.00	7,880.00	3,900
DL	0.70	0.20	0.01	0.02	0.02	0.20	0.01	0.01	0.01	0.01	0.01	0.01	0.01	2.00	10

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	500.00	380.00	0.00	0.50	91.00	688.00								7,150.00	3,320
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	65.10	33.00	0.00	0.30	24.60	133.00								1,640.00	962
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	780.00	493.00	0.00	0.90	92.50	822.00								5,870.00	4,470
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

C. FRESH WATER ANALYSES

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.40	0.00	6.40								284.00	0
DL	0.01	0.01	0.01								0.01	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								475.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

IW Mizner

Sample description: IW Mizner

Sample date: 08/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	34.90	0.00	34.90								806.00	
DL	0.01	0.01	0.01								10.00	

TID Well

Sample description: TID Well

Sample date: 12/07/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.80	0.00	12.80								1,510.00	0
DL	0.01	0.01	0.01								10.00	1

River

River

Sample description: River

Sample date: 07/21/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.40	0.00	1.40								467.00	268
DL	1.00	0.01	1.00								1.00	1

D. SOIL ANALYSES

Armelin

Armelin

Armelim Soil

Sample and source description: Armelim Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			59.00				
DL			1.00				

Home

Dairy Soil

Sample and source description: Dairy Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			1.00				

Horseranch

Horse Ranch Soil

Sample and source description: Horse Ranch Soil

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			39.00				
DL			1.00				

Vineyard

Vineyard

Vineyard Soil

Sample and source description: Vineyard Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			153.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

Armelin - 11/05/2022: Oats, silage-soft dough

Armelin Oats

Sample and source description: Armelin Oats

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	3,600.00	18,000.00		6.60
DL	0.01	0.01	0.01		1.00

Armelin - 05/21/2023: Corn, silage

Armelin Corn

Sample and source description: Armelin Corn

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	4,100.00	19,100.00		7.70
DL	0.01	0.01	0.01		1.00

Denair - 10/05/2020: Alfalfa, hay

Denair Alfalfa First Cut

Sample and source description: Denair Alfalfa First Cut

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 23.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,300.00	3,200.00	31,400.00		12.40
DL	0.01	0.01	0.01		1.00

Denair Alfalfa 2nd cut

Sample and source description: Denair Alfalfa 2nd cut

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,100.00	3,500.00	25,800.00		10.00
DL	0.01	0.01	0.01		1.00

Home - 11/18/2022: Oats, silage-soft dough

Home Ranch Oats

Sample and source description: Home Ranch Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,600.00	3,300.00	42,100.00		13.60
DL	0.01	0.01	0.01		1.00

Home - 05/21/2023: Corn, silage

Home - 05/21/2023: Corn, silage

Home Ranch Corn

Sample and source description: Home Ranch Corn

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,400.00	21,100.00		7.80
DL	0.01	0.01	0.01		1.00

Horseranch - 11/17/2022: Oats, silage-soft dough

Horse Ranch Oats

Sample and source description: Horse Ranch Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,500.00	4,800.00	49,600.00		17.10
DL	0.01	0.01	0.01		1.00

Horseranch - 05/21/2023: Corn, silage

Horse Ranch Corn

Sample and source description: Horse Ranch Corn

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	3,000.00	15,900.00		6.30
DL	0.01	0.01	0.01		1.00

River - 11/19/2022: Oats, silage-soft dough

River Oats

Sample and source description: River Oats

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,300.00	15,200.00		7.80
DL	0.01	0.01	0.01		1.00

River - 06/21/2023: Corn, silage

River Corn

Sample and source description: River Corn

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	0.30	17,900.00		6.30
DL	0.01	0.01	0.01		1.00

Vineyard - 11/16/2022: Oats, silage-soft dough

Vineyard Oats

Sample and source description: Vineyard Oats

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,500.00	4,600.00	51,600.00		14.50
DL	0.01	0.01	0.01		1.00

Vineyard - 05/21/2023: Corn, silage

Vineyard Corn

Sample and source description: Vineyard Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	3,400.00	23,600.00		8.20
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

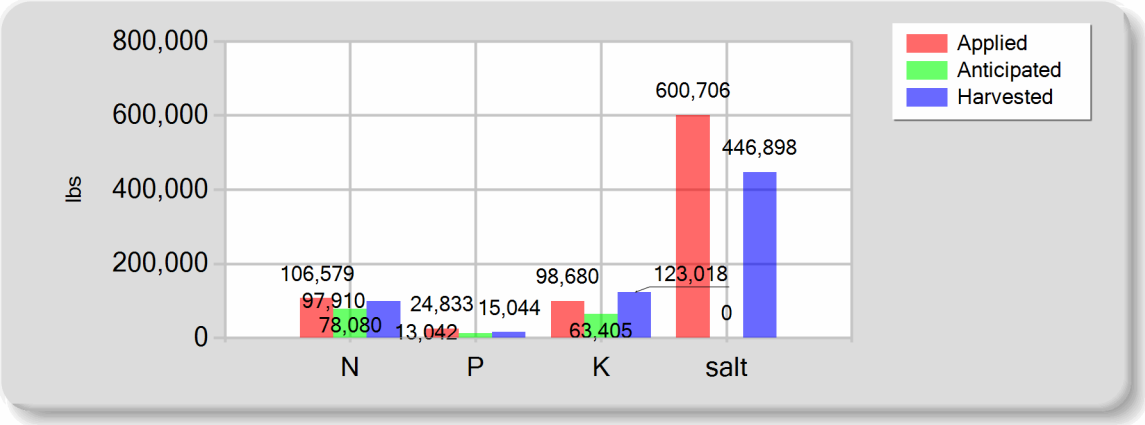
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

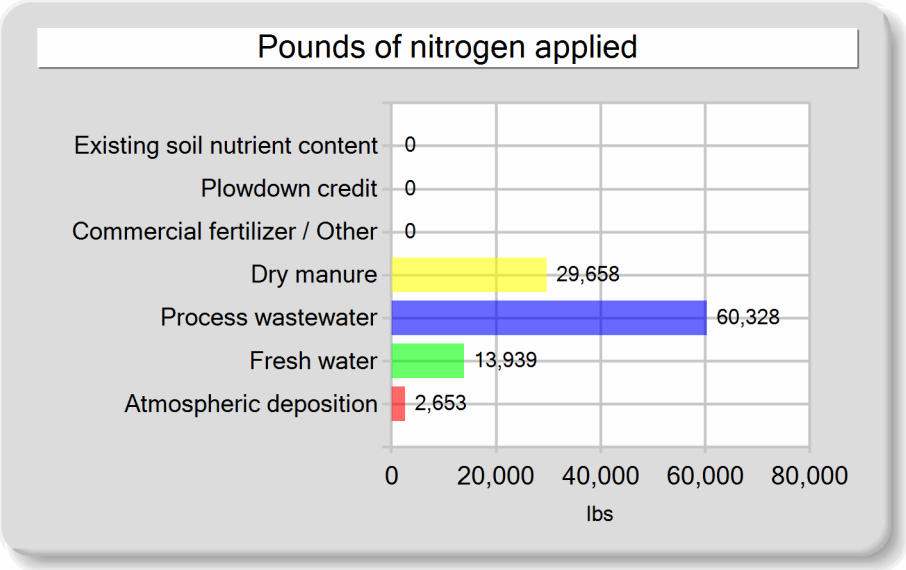
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

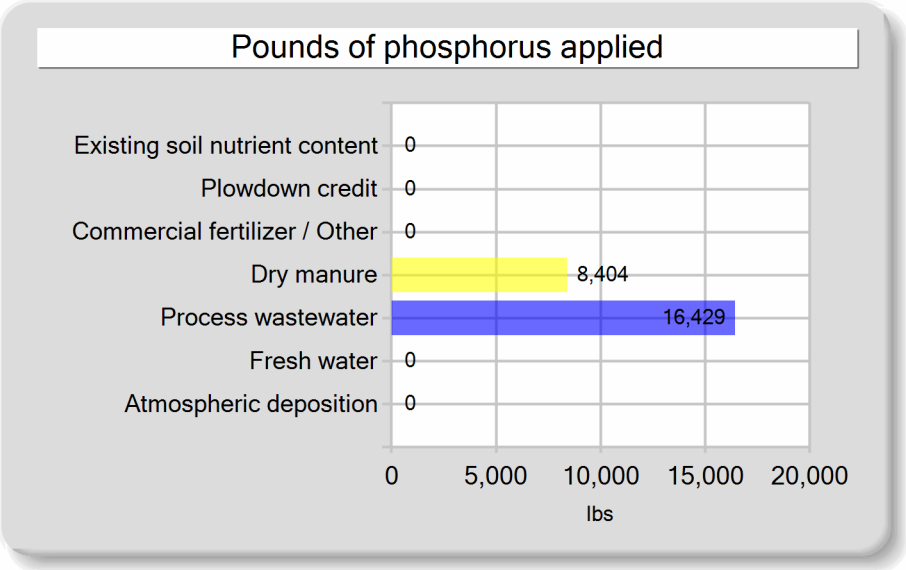
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	29,658.17	8,403.95	5,963.47	0.00
Process wastewater	60,328.07	16,428.87	92,716.12	513,537.45
Fresh water	13,939.49	0.00	0.00	87,168.26
Atmospheric deposition	2,653.00	0.00	0.00	0.00
Total nutrients applied	106,578.73	24,832.82	98,679.59	600,705.71
Anticipated crop nutrient removal	78,080.00	13,042.40	63,404.80	0.00
Actual crop nutrient removal	97,909.70	15,044.02	123,018.32	446,897.94
Nutrient balance	8,669.03	9,788.80	-24,338.73	153,807.77
Applied to removed ratio	1.09	1.65	0.80	1.34

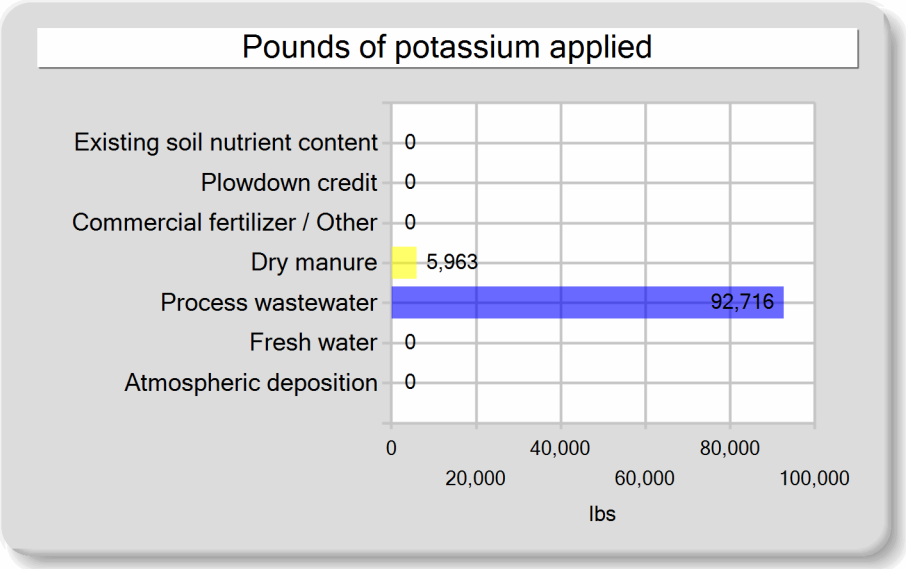
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

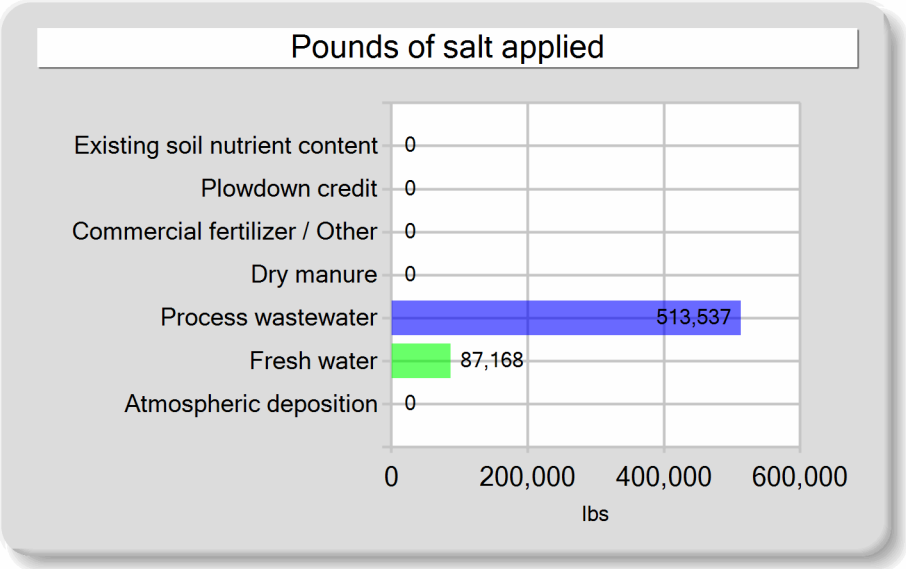


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Manure excreted by herd and nitrogen has been modified to reflect recent studies taken from herds utilizing the same feed supplement being utilized on our facility. A 33% reduction of excreted was used for reporting purposes.

Denair alfalfa field had 3 cuts of alfalfa haylage and 3 cuts of baled hay, which is why yields are high.

Horse ranch field had above average yield and sample was higher NPK concentration than normal which creates a higher than normal removal rate

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Tiberio Cardoso	Tiberio Cardoso
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
M & M Cardoso and Sons LTD	Armelim or Tiberio Cardoso
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 18, 2024	Jun 18, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Tiberio Cardoso

Name of Dairy Facility: M & M Cardoso & Sons Dairy LTD

Facility Address:

Number and Street	City	County	Zip Code
16489 August AVE	Delhi	Merced	95315

Contact Person Name and Phone Number:

Name	Phone Number
Tiberio Cardoso	(209) 648-4032

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Cardoso Farm Services Inc.

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 60	Hilmar	CA	95324

Contact Person:

Name	Phone Number
Arlindo Cardoso	(209) 564-6329

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Joey Machado	(209) 606-8475

Address	City	State	Zip Code
111 Youngstown Rd	Turlock	CA	95380

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Turlock	95380

Street and nearest cross street (if no address)	County
Youngstown Rd	Stanislaus

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 11/14/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,274.00 tons

Manure Solids Content: 74.0 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Tiberio Cardoso — Jun 18, 2024

Operator Signature — Date

Arlindo Cardoso — Jun 13, 2024

Hauler Signature — Date

DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 08/03/2023 15:09
Reported: 08/07/2023 11:59

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0567-01	IW Mizner	Well Water	M.Dekeyrel		08/02/2023 11:00

Default Cooler Temperature on Receipt °C: -3.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 08/03/2023 15:09
Reported: 08/07/2023 11:59

Sample Results

Sample: IW Mizner
23H0567-01 (Water)

Sampled: 8/2/2023 11:00
Sampled By: M.Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.81**	mmhos/cm	0.01	1		08/04/23 15:07	SM 2510 B		BEH0247
Electrical Conductivity umhos	**806**	umhos/cm	10.0	1		08/04/23 15:07	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 11:00	Field		BEH0308
Nitrate Nitrogen as NO3N	**34.9**	mg/L	0.1	1	10	08/03/23 21:14	EPA 300.0		BEH0186
pH	**7.0**	units	1.0	1		08/04/23 15:07	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:07	SM 2510 B		BEH0247

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Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 08/03/2023 15:09
Reported: 08/07/2023 11:59

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0186										
Blank (BEH0186-BLK1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0186-BLK2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0186-BS1)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		93.5	90-110		
Duplicate (BEH0186-DUP1)		**Source: 23H0558-01**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	0.07		0.1	mg/L		0.07			1.48	10
Matrix Spike (BEH0186-MS1)		**Source: 23H0558-01**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.07	92.2	90-110		
Reference (BEH0186-SRM1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.7	90-110		

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DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 08/03/2023 15:09
Reported: 08/07/2023 11:59

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247										
Blank (BEH0247-BLK1)					Prepared & Analyzed: 8/4/2023					
pH	5.4		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK2)					Prepared & Analyzed: 8/4/2023					
pH	7.5		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0247-DUP1)		**Source: 23H0569-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1.52		0.01	mmhos/cm		1.54			1.70	10
pH	7.5		1.0	units		7.5			0.133	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1540			1.70	10
Duplicate (BEH0247-DUP2)		**Source: 23H0612-01**			Prepared & Analyzed: 8/4/2023					
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity	0.14		0.01	mmhos/cm		0.14			0.0713	10
Electrical Conductivity umhos	140		10.0	umhos/cm		140			0.0712	10
Reference (BEH0247-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	544			umhos/cm	538.0		101	90-110		
Reference (BEH0247-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.9			units	7.790		101	.7163-101.28		
Reference (BEH0247-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0247-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 08/03/2023 15:09
Reported: 08/07/2023 11:59

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247 (Continued)										
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0247-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0247-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH0247-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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08/03/23 15:09 23H0567

JG

WATER WORK REQUEST

Acct No. / Cons.

Bill To: 21351 | 8

Purchase Order No. _____ Results Needed By _____

Client: Cardoso & Sons Dairy LP, M & M
Address: 16489 August Ave
City, State, Zip: Delhi, CA 95315
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Tiberio Cardoso - (209) 648-4032

Ranch _____

Date sampled: 8/2/23

Sampled by: M Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

No. of Samples 1 No. Bottles 3

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. IW Mizner	Sampled From:	8/2/23	11AM		-3.8
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MDekeyrel	CAS Inc	8/2/23 11AM	8/3/23 1pm
Second	JLR	DLV	8-3-23 1:00	8-3-23 2:30
Third				
Fourth	JG	DLV	8/3/23 15:09	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic	+	JG	8/3/23							
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1									
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

08/03/23 15:09 23H0567

SEND OUT

DELLAVALLE™ LABORATORY INC

Report of Dairy Well Water Analysis

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi CA 95315
00-0021351 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23H0567
Sampled By: M.Dekeyrel
Requested By: Tiberio Cardoso
Submitted Date: 08/03/23
Reported Date: 08/07/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	IW Mizner	8/2/23	11:00	806	0.81	34.9	*		7

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi CA 95315
00-0021351 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23H0567
Sampled By: M.Dekeyrel
Requested By: Tiberio Cardoso
Submitted Date: 08/03/23
Reported Date: 08/07/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	IW Mizner	8/2/23	11:00	806	0.81	**34.9**	95.28	7.94

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi CA 95315
00-0021351 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0451
Sampled By: M.Dekeyrel
Requested By: Tiberio Cardoso
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW MB	12/06/23	12:40	475	0.48	0.7	ND		7.9
2	TID Well	12/06/23	12:55	1510	1.51	12.8	ND		7.3

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi CA 95315
00-0021351 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0451
Sampled By: M.Dekeyrel
Requested By: Tiberio Cardoso
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project:
Crop ID:

		Date Sampled	Time Sampled	**EC** µmhos/cm	**EC** mmhos/cm	**NO_3-N** mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW MB	12/06/23	12:40	475	0.48	0.7	1.91	0.16
2	TID Well	12/06/23	12:55	**1510**	**1.51**	**12.8**	34.94	2.91

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0451-01	DW MB	Ag Water	M.Dekeyrel		12/06/2023 12:40
23L0451-02	TID Well	Ag Water	M.Dekeyrel		12/06/2023 12:55

Default Cooler Temperature on Receipt °C: -2.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Sample Results

Sample: DW MB
23L0451-01 (Water)

Sampled: 12/6/2023 12:40
Sampled By: M.Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.48**	mmhos/cm	0.01	1		12/11/23 16:09	SM 2510 B		BEL0301
Electrical Conductivity umhos	**475**	umhos/cm	10.0	1		12/11/23 16:09	SM 2510 B		BEL0301
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 12:40	Field		BEL0480
Nitrate Nitrogen as NO3N	**0.7**	mg/L	0.1	1	10	12/08/23 06:14	EPA 300.0		BEL0227
Temperature	**25.0**	units	0.0	1		12/11/23 16:09	SM 4500-H+	H	BEL0301
pH	**7.9**	units	1.0	1		12/11/23 16:09	SM 4500-H+	H	BEL0301

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DELLAVALLE™
LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Sample Results
(Continued)

Sample: TID Well
23L0451-02 (Water)

Sampled: 12/6/2023 12:55
Sampled By: M.Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.51**	mmhos/cm	0.01	1		12/11/23 16:17	SM 2510 B		BEL0301
Electrical Conductivity umhos	**1510**	umhos/cm	10.0	1		12/11/23 16:17	SM 2510 B		BEL0301
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 12:55	Field		BEL0480
Nitrate Nitrogen as NO3N	**12.8**	mg/L	0.1	1	10	12/08/23 06:34	EPA 300.0		BEL0227
Temperature	**25.0**	units	0.0	1		12/11/23 16:17	SM 4500-H+	H	BEL0301
pH	**7.3**	units	1.0	1		12/11/23 16:17	SM 4500-H+	H	BEL0301

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DELLAVALLE™ LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0227										
Blank (BEL0227-BLK1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK4)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0227-BS1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		106	90-110		
LCS (BEL0227-BS2)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0227-BS3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
Duplicate (BEL0227-DUP1)		**Source: 23L0251-01**			Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	2.6		0.1	mg/L		2.6			0.610	10
Duplicate (BEL0227-DUP2)		**Source: 23L0351-02**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			0.00	10
Duplicate (BEL0227-DUP3)		**Source: 23L0446-01**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	3.9		0.1	mg/L		3.9			0.00	10
Matrix Spike (BEL0227-MS1)		**Source: 23L0251-01**			Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	7.8		0.1	mg/L	5.000	2.6	103	90-110		
Matrix Spike (BEL0227-MS2)		**Source: 23L0351-02**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.04	105	90-110		
Matrix Spike (BEL0227-MS3)		**Source: 23L0446-01**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	9.1		0.1	mg/L	5.000	3.9	104	90-110		
Reference (BEL0227-SRM1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0227-SRM2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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DELLAVALLE™ LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0227 (Continued)										
Reference (BEL0227-SRM3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0227-SRM4)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0301										
Blank (BEL0301-BLK1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0301-BLK2)					Prepared & Analyzed: 12/11/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0301-BLK3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.3		1.0	units						
Duplicate (BEL0301-DUP1)		**Source: 23L0451-01**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.48		0.01	mmhos/cm		0.48			0.0421	10
Electrical Conductivity umhos	476		10.0	umhos/cm		475			0.0421	10
pH	7.9		1.0	units		7.9			0.381	10
Duplicate (BEL0301-DUP2)		**Source: 23L0453-02**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.80		0.01	mmhos/cm		0.79			0.126	10
Electrical Conductivity umhos	795		10.0	umhos/cm		794			0.126	10
pH	7.8		1.0	units		7.8			0.129	10
Reference (BEL0301-SRM1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	437			umhos/cm	426.0		103	90-110		
Reference (BEL0301-SRM2)					Prepared & Analyzed: 12/11/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0301-SRM3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0301-SRM4)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0301-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

M & M Cardoso & Sons Dairy LP
16489 August Ave
Delhi, CA 95315

Account# 00-0021351
Account Manager: Ben Nydam
Submitted By: Tiberio Cardoso

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:07

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0301 (Continued)										
Reference (BEL0301-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0301-SRM6)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0301-SRM7)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0301-SRM8)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		99.8	97.5-102.5		

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12/07/23 15:35 23L0451

WATER WORK REQUEST

Bill To: Acct No. 21351 Cons. 8

Purchase Order No. Results Needed By

Client: Cardoso & Sons Dairy LP, M & M
Address: 16489 August Ave
City, State, Zip: Delhi, CA 95315
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Tiberio Cardoso - (209) 648-4032

Ranch:

Date sampled: 12/6/23

Sampled by: mDekeuvel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

No. of Samples 2 No. Bottles 2

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(Y) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW MB	Sampled From:	12/6/23	1240p	Ø	-2.9
2. TID Well	Sampled From:	12/6/23	1255p	Ø	-2.9
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MDekeuvel	OAS Inc	12/6/23 1255p	12/7/23 11A
Second	JLF	DLV	12-7-23 11:00	12-7-23 3:15
Third				
Fourth		DLI	12-7-23 15:35	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___

Shipping
$ ___ In
$ ___ Out
Date ___

Signature ___

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

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	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1								
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT